ENCODE × ARC - PROGRAMMABLE MONEY HACKATHON · FINAL SUBMISSION

Parley

Agents that negotiate the price, then pay it.

A buyer and a seller strike deals inside owner-set guardrails and settle in USDC on Arc.

Track: Agentic Economy

github.com/edwardparker8760/parley

88 tests green

THE GAP

Agent payments exist. Agent *pricing* does not.

Today: fixed-price payments

Existing agentic-payment demos, including Circle's own nanopayments starter, show an agent paying a **fixed, posted price**. The economically interesting step, deciding *what the price should be*, never happens.

Parley: price discovery

Two agents with **opposing interests** negotiate price, quantity and terms on behalf of their owners. Settlement is the **last step, not the product**.

> *If agents are going to transact for us, they must be able to disagree, concede, and walk away, without ever breaking their owner's limits.*

THE CLAIM

Owner limits are arithmetic, not instructions

Deterministic layer: correctness

- Utility functions and a back-loaded concession schedule
- ZOPA inferred from revealed offers only
- **Hard guardrail clamp** computes the feasible band; nothing outside it can be sent
- An **independent egress guard** on the bus re-derives the band and rejects breaches

LLM layer: judgement and legibility

- Chooses *where inside a 2% window* to land
- Writes the rationale attached to every offer
- Schema-validated; an out-of-band pick is discarded, its words kept
- Timeout, transport error and bad schema all fall back deterministically

> *The band is computed **before** the model is consulted and re-checked **twice** after. A fully compromised model still cannot produce an out-of-band offer.*

TESTED, NOT ASSERTED

The safety claim, mechanically checked

0

out-of-band offers on the wire from a model answering **99999999** to every prompt, across all three scenarios. Every refusal is logged with the number arithmetic threw away.

0

settlement calls on a walk-away. Unreachable by construction, asserted by a counting-spy test over scenario C, against exactly 1 for scenario A.

88

tests: band properties over an adversarial corpus, prompt injection through the counterparty's own text, prompt-leak in both directions, byte-identical replay from SQLite.

> *Swapping the LLM provider mid-build changed **zero** tests in the injection suite. The safety property belongs to the arithmetic, not to the model.*

CIRCLE STACK

What each Circle tool does here

Arc Testnet

`eip155:5042002`. The chain settlement lands on. USDC at `0x3600...`, RPC `rpc.testnet.arc.network`.

Circle Gateway

`GatewayClient` holds the buyer's Gateway balance and signs the EIP-3009 authorisation. Deposit, balances and withdrawal all run through it.

Nanopayments x402

`@circle-fin/x402-batching@3.2.0`. Buyer `pay()` runs the full 402 flow against a seller endpoint built with `createGatewayMiddleware`.

x402 facilitator

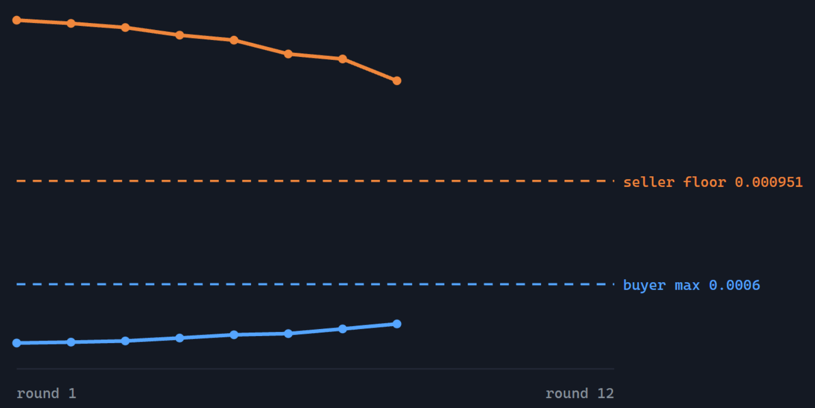
`gateway-api-testnet.circle.com` verifies and settles. Passed explicitly, because the SDK default is mainnet.

> *Read from the installed SDK, not the blog post: `GatewayClient` takes a raw EVM key via viem. No API key, no entity secret, no Developer-Controlled Wallets in the payment path. That finding made the credential blocker smaller, and it is written down.*

SCENARIO C, THE ONE THAT MATTERS

No overlap: both walk away, nothing is paid

CONVERGENCE dashed lines are each side's private limit, shown to the audience only



No overlap exists. seller floor 951 exceeds buyer ceiling 600: no price satisfies both owners

WALK-AWAY no payment was made

A deal was never possible: the two owners' limits do not overlap.

BUYER

NO_ZOPA_PRICE

bound by: buyer maximum unit price and total spend cap

final gap 0.000827/call

after 9 rounds

buyer maximum unit price and total spend cap bound after 9 rounds, final gap 827 micro-USDC

SELLER

NO_ZOPA_PRICE

bound by: seller margin floor over cost basis

final gap 0.000827/call

after 9 rounds

seller margin floor over cost basis bound after 9 rounds, final gap 827 micro-USDC

The dashed lines are each side's private limit: neither agent sees the other's, the audience sees both. The seller's floor sits **above** the buyer's ceiling, so no price satisfies both owners. The agents establish that in nine rounds, then each files a post-mortem naming the limit that bound it.

WHERE IT LANDED

Built, and what is deliberately not claimed

Built

- ✓ Protocol, ledger and a turn loop whose round cap sits outside both agents
- ✓ Guardrail engine: two independent checks over one pure band function
- ✓ Utility, concession schedule, dual ZOPA detection
- ✓ Bounded LLM, wired in, every consultation logged
- ✓ Settlement on accept, post-mortems on walk-away
- ✓ One-screen dashboard, live and replayable

Not claimed

- Settlement runs on the **local stub**. The Arc adapter is implemented and tested offline, but no wallet is funded, so no real money has moved and every receipt is badged **SIMULATED**.
- Testnet only.
- Prompts go to a third-party API.
- No identity or reputation layer. Scoped, then cut on day one of six.

> *Next: on-chain, sybil-resistant reputation, so a seller's history survives across marketplaces and cannot be reset by spinning up a fresh address.*